



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Low Power Digital Integrated Circuits
Course Number:	ELE734
Semester/Year (e.g. F2016)	F2025

Instructor:	Dr. Andy Gean Ye
Lab Report Number	Lab #2 - Lab
Lab Title	Lab 2 - Lab: CMOS Inverter

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Lab 2 : Pre - Lab:

1)

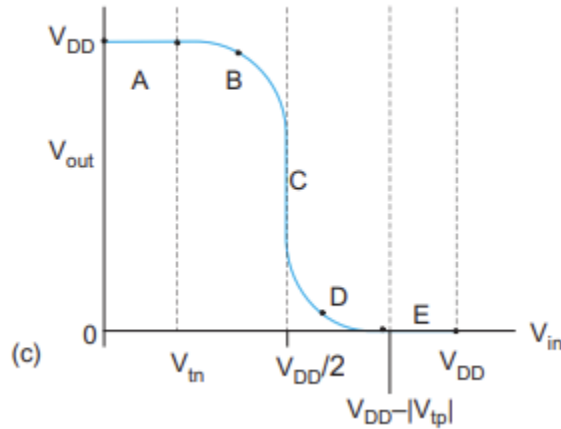


Figure 1: CMOS Inverter DC Characteristics in the form of V_{in} vs V_{out} ($V_{in} = V_{gs}$, $V_{out} = V_{ds}$)

TABLE 2.3 Summary of CMOS inverter operation

Region	Condition	p-device	n-device	Output
A	$0 \leq V_{in} < V_{tn}$	linear	cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \leq V_{in} < V_{DD}/2$	linear	saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	saturated	saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{tp} $	saturated	linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - V_{tp} $	cutoff	linear	$V_{out} = 0$

Table 2: CMOS Inverter Characteristics(V_{in} vs V_{out}) graph; transistor operation region for operating points

In a usual case, the $\beta_p = \beta_n$, and the V_{inv} remains $V_{DD}/2$ for an inverter. However, the beta ratio which is determined by the equation: $r = \beta_p/\beta_n$, can be modified and changed to increase or decrease, and this concept is assigned to the term skewed inverters. If the beta ratio increases ($r > 1$), then the inverter is HI-skewed, implying that $\beta_p > \beta_n$. This type of inverter has a stronger PMOS transistor that can hold and keep the PMOS turned on for a longer time (shifts DC characteristic graph more to the right). This ultimately increases the V_{inv} (Transition period/input threshold for input change; PMOS to NMOS) to greater than $V_{DD}/2$, as PMOS remains active for a greater period. Now if beta ratio decreases ($r < 1$), a LO-skewed inverter is made, which has a weaker PMOS transistor (shifts DC characteristic graph more to the left), thus

the PMOS does not stay active and on for a long time period, ultimately reducing the transition period/switching threshold.

2)

The output voltage equals the input voltage at the inverter threshold. On top of that we know the PMOS and NMOS are both conducting in saturation mode. They will have the same node.

The relationship between the drain current and the inverter threshold can be shown through the following formula:

$$\begin{aligned} I_{dn} &= \frac{\beta_n}{2} (V_{inv} - V_{tn})^2 \\ I_{dp} &= \frac{\beta_p}{2} (V_{inv} - V_{DD} - V_{tp})^2 \end{aligned} \quad (2.54)$$

In this case we set the currents as the same (equal and opposite signs) so we can solve for V_{inv} as a function of r :

$$V_{inv} = \frac{V_{DD} + V_{tp} + V_{tn} \sqrt{\frac{1}{r}}}{1 + \sqrt{\frac{1}{r}}} \quad (2.55)$$

If the transistors are velocity saturated (short-channel devices), we can use the equations:

$$\begin{aligned} I_{dn} &= W_n C_{ox} v_{sat-n} (V_{inv} - V_{tn}) \\ I_{dp} &= W_p C_{ox} v_{sat-p} (V_{inv} - V_{DD} - V_{tp}) \end{aligned} \quad (2.56)$$

Similarly like before, we make both currents equal and opposite to one another and equate the formula to solve for V_{inv} :

$$V_{inv} = \frac{V_{DD} + V_{tp} + V_{tn} \frac{1}{r}}{1 + \frac{1}{r}} \quad (2.57)$$

3)

Noise margin is the parameter that controls and determines the allowable noise voltage at the input of the gate, allowing the output to be compatible and not corrupted. Now, as the supply voltage decreases so those the noise margin decrease:

$$NM_L = V_{IL} - V_{OL} \quad (2.58)$$

$$NM_H = V_{OH} - V_{IH} \quad (2.59)$$

4)

$$\tau_{pdr} = \ln 2 * R_{eqp} * C_L$$

$$\tau_{pdf} = \ln 2 * R_{eqn} * C_L$$

$$R_{eq} = \frac{V_{DD}}{2 * \ln 2 * I_{Dsat\ n,p}}$$

We know that the overall propagation delay of the inverter is defined as the average of the output rising delay and the output falling delay:

$$\tau_{TP} = \frac{\tau_{pdr} + \tau_{pdf}}{2}$$

Substitute, $I_{Dsat\ n,p}$ with the Saturated MOSFET current formula: $I = \frac{\beta}{2} (V_{gs} - V_{th})^2$

Also note that, $\beta = \mu * C_{ox} * \frac{W}{L}$

$$R_{eqn} = \frac{V_{DD}}{2 * \ln 2 * I_{Dsat\ n}} = \frac{V_{DD}}{2 * \ln 2 * \frac{\beta_n}{2} (V_{gs} - V_{th,n})^2} = \frac{V_{DD}}{\ln 2 * \mu_n * C_{ox,n} * \frac{W_n}{L} (V_{gs} - V_{th,n})^2}$$

$$R_{eqp} = \frac{V_{DD}}{2 * \ln 2 * I_{Dsat\ p}} = \frac{V_{DD}}{2 * \ln 2 * \frac{\beta_p}{2} (V_{sg} - V_{th,p})^2} = \frac{V_{DD}}{\ln 2 * \mu_p * C_{ox,p} * \frac{W_p}{L} (V_{sg} - V_{th,p})^2}$$

After finding the equation for R_{eqp} , substitute it back in the τ_{pdf} and τ_{pdr} formulas:

$$\tau_{pdf} = \ln 2 * R_{eqn} * C_L = \ln 2 * \frac{V_{DD}}{\ln 2 * \mu_n * C_{ox,n} * \frac{W_n}{L} (V_{gs} - V_{th,n})^2} * C_L = \frac{V_{DD} * C_L}{\mu_n * C_{ox,n} * \frac{W_n}{L} (V_{gs} - V_{th,n})^2}$$

$$\tau_{pdr} = \ln 2 * R_{eqp} * C_L = \ln 2 * \frac{V_{DD}}{\ln 2 * \mu_p * C_{ox,p} * \frac{W_p}{L} (V_{sg} - V_{th,p})^2} * C_L = \frac{V_{DD} * C_L}{\mu_p * C_{ox,p} * \frac{W_p}{L} (V_{sg} - V_{th,p})^2}$$

Lastly, use the overall propagation formula for average delay in output falling and output rising:

$$\tau_{TP} = \frac{\tau_{pdr} + \tau_{pdf}}{2} = \frac{V_{DD} * C_L}{2} \left(\frac{1}{\mu_n * C_{ox,n} * \frac{W_n}{L} (V_{gs} - V_{th,n})^2} + \frac{1}{\mu_p * C_{ox,p} * \frac{W_p}{L} (V_{sg} - V_{th,p})^2} \right)$$

5)

Compute the Gate Oxide Capacitance per Unit Area:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 * 8.854 * 10^{-12}}{4.09 * 10^{-9}} = 0.008463 = 8.463 * 10^{-3} F/m^2$$

Compute the Required Drain Current needed for both NMOS and PMOS; since delay for both transistors(rise and fall delays) are the same(100 ps):

$$\tau_P = \frac{V_{DD} * C_L}{I_{Dsat}} \Rightarrow I_{Dsat\ n,p} = \frac{V_{DD} * C_L}{\tau_P} = \frac{1.8 * (50 * 10^{-15})}{100 * 10^{-12}} = 9.0 * 10^{-4} A$$

Compute the width for the NMOS:

$$I_{Dsat, n} = \frac{1}{2} \mu_n * C_{ox, n} * \frac{W_n}{L} (V_{gs} - V_{th, n})^2$$

$$\frac{W_n}{L} = \frac{2 * I_{Dsat, n}}{\mu_n * C_{ox, n} * (V_{DD} - V_{tn})^2} = \frac{2(9 * 10^{-4})}{0.0459 * (8.463 * 10^{-3}) * (1.8 - 0.445)^2} = 2.52$$

$$\frac{W_n}{L} = 2.52 \Rightarrow W_n = 2.52(180nm) = 454 nm = 0.454 \mu m$$

Compute the width for the PMOS:

$$I_{Dsat, p} = \frac{1}{2} \mu_p * C_{ox, p} * \frac{W_p}{L} (V_{sg} - V_{th, p})^2$$

$$\frac{W_p}{L} = \frac{2 * I_{Dsat, p}}{\mu_p * C_{ox, p} * (V_{DD} - |V_{tp}|)^2} = \frac{2(9 * 10^{-4})}{0.0109 * (8.463 * 10^{-3}) * (1.8 - 0.438)^2} = 2.52$$

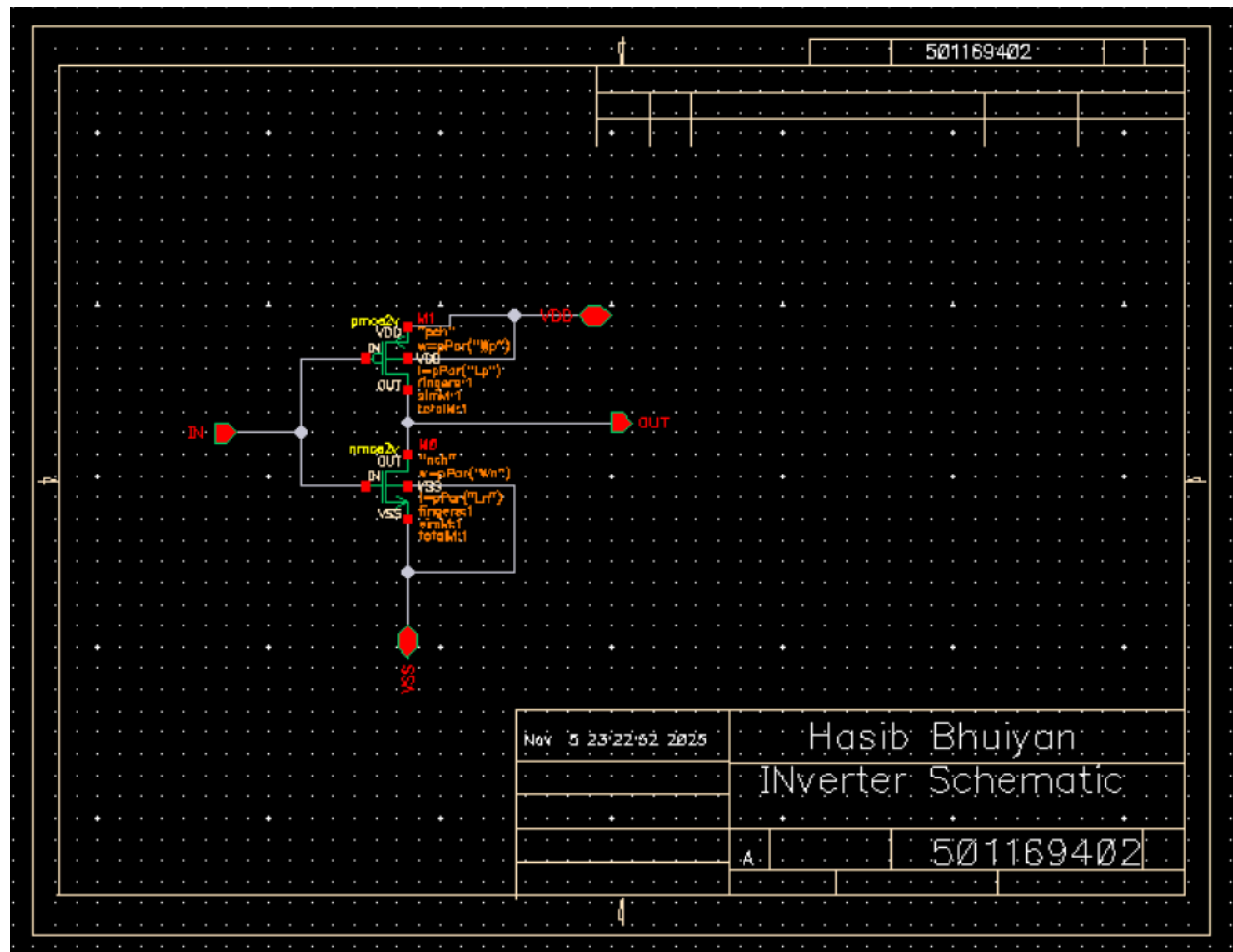
$$\frac{W_p}{L} = 10.52 \Rightarrow W_p = 10.52(180nm) = 1893.6 nm = 1.89 \mu m$$

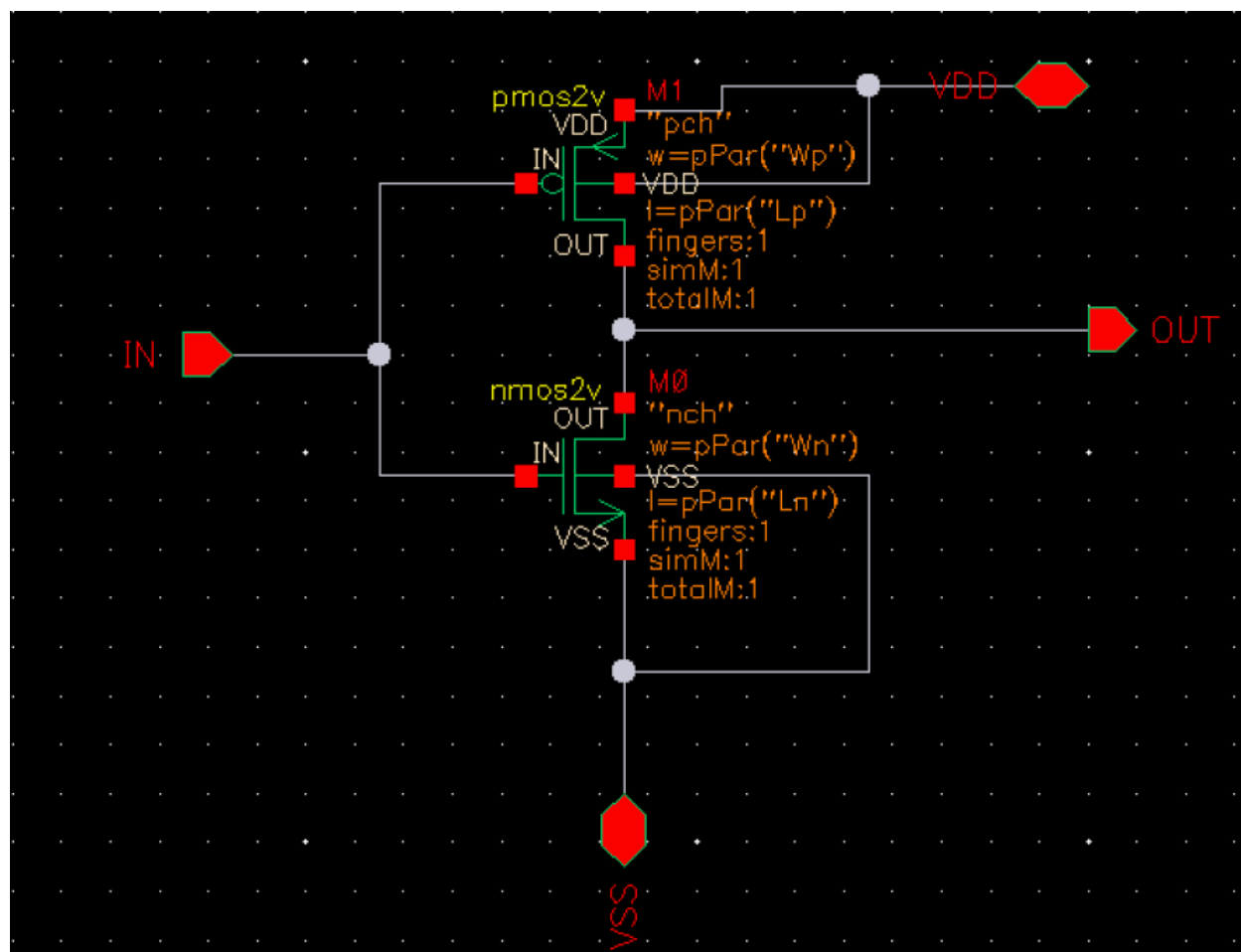
Ratio r:

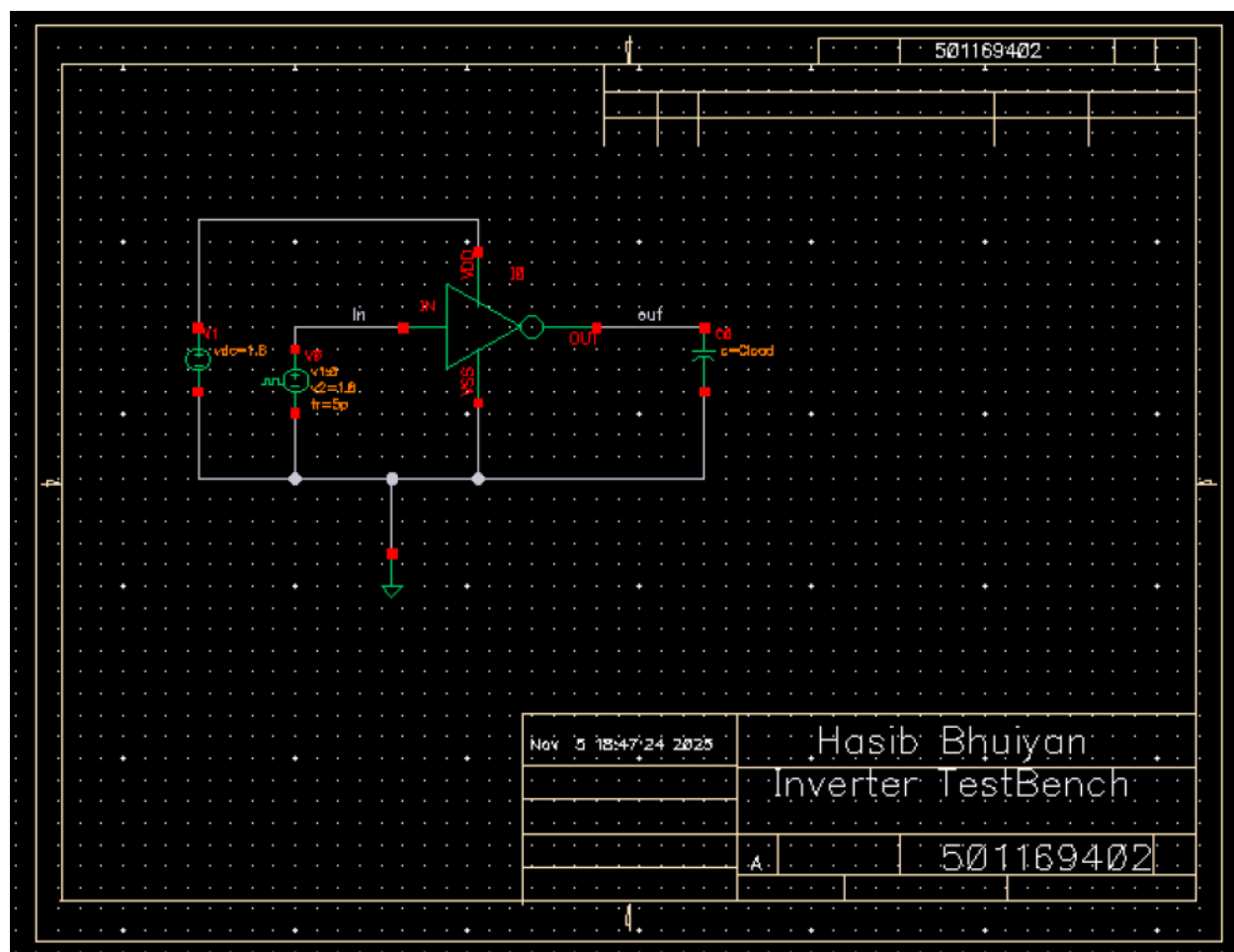
$$r = \frac{\beta_p}{\beta_n} = \frac{\mu_p (\frac{W_n}{L})}{\mu_n (\frac{W_p}{L})} = \frac{0.0109 * 10.52}{0.0459 * 2.52} = 0.98$$

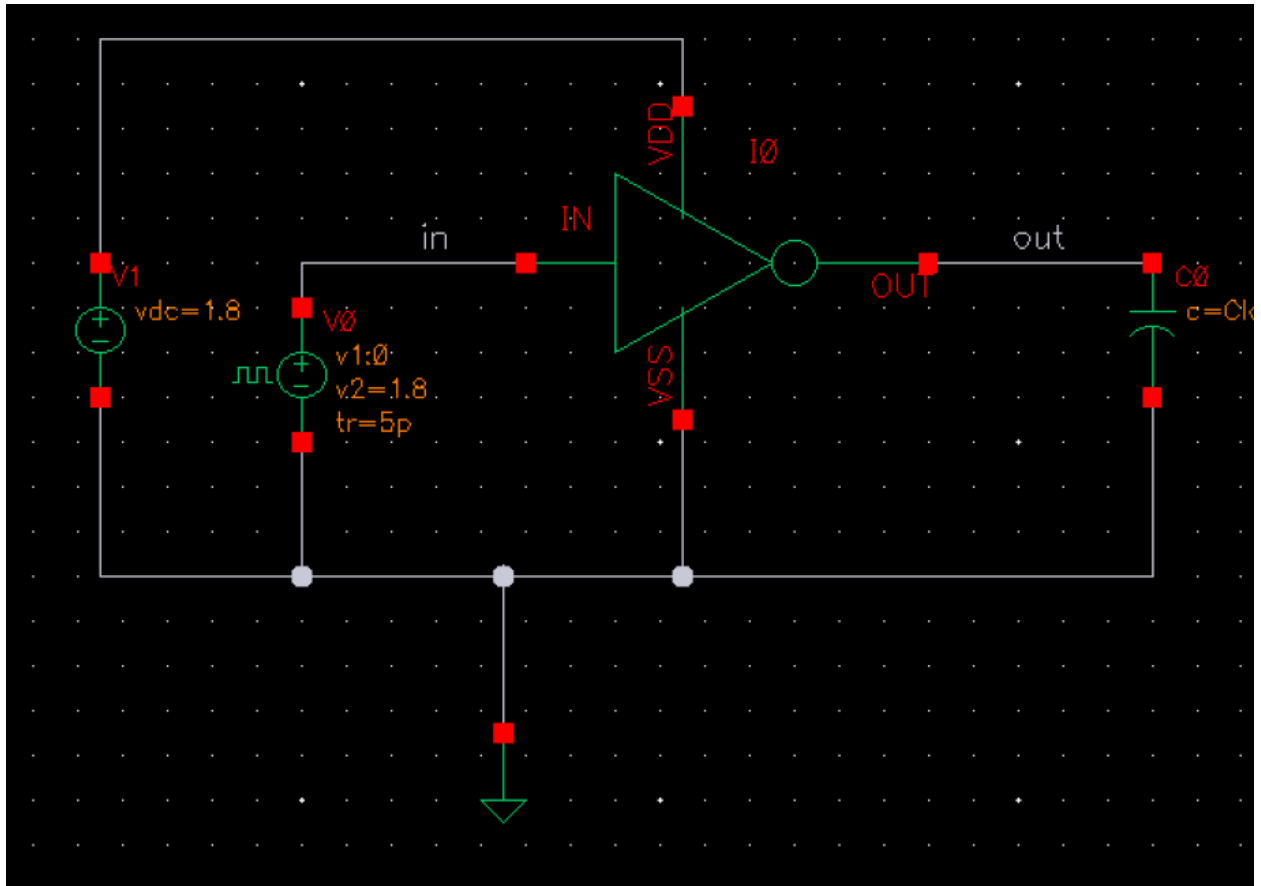
Thus, from the beta ratio r, it is seen that the r value is just slightly below 1; this shows that the CMOS inverter is of a LO-skewed type(just slightly), implying that the PMOS is weaker and has a transition a bit earlier, with the switching threshold earlier too. Also note that from the ratio r calculation we can see that NMOS has higher mobility(electron mobility) than the PMOS(hole mobility) as usual; due to this significant gap, the PMOS is sized larger just to compensate for the lower hole mobility, ensuring that r stays near 1 and ultimately the NMOS and PMOS have around an equal “turn-on” time(symmetrical rise/fall times too).

Lab 2 : In - Lab:







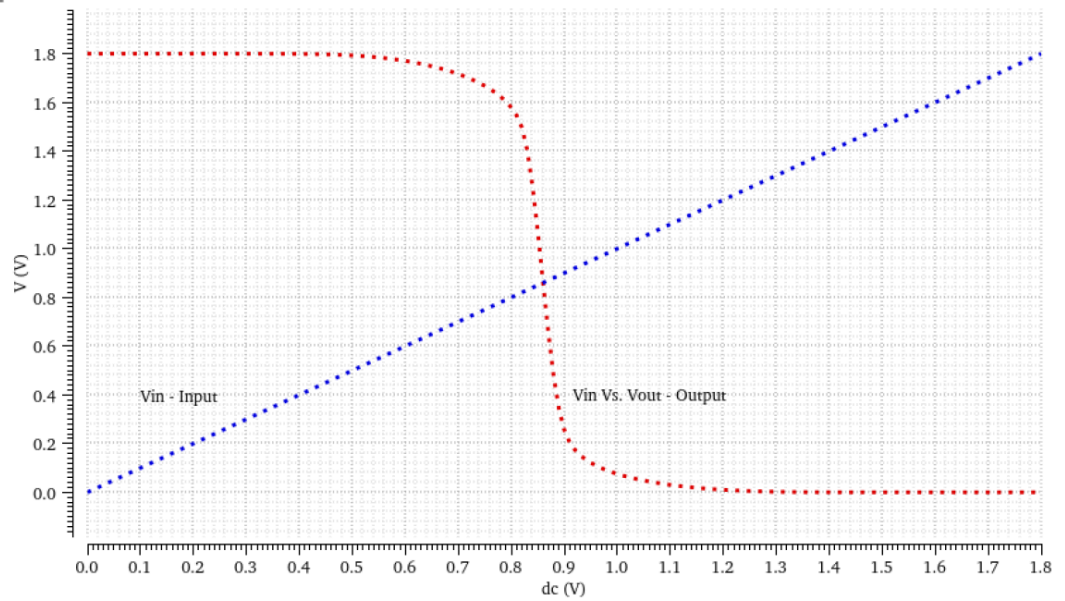


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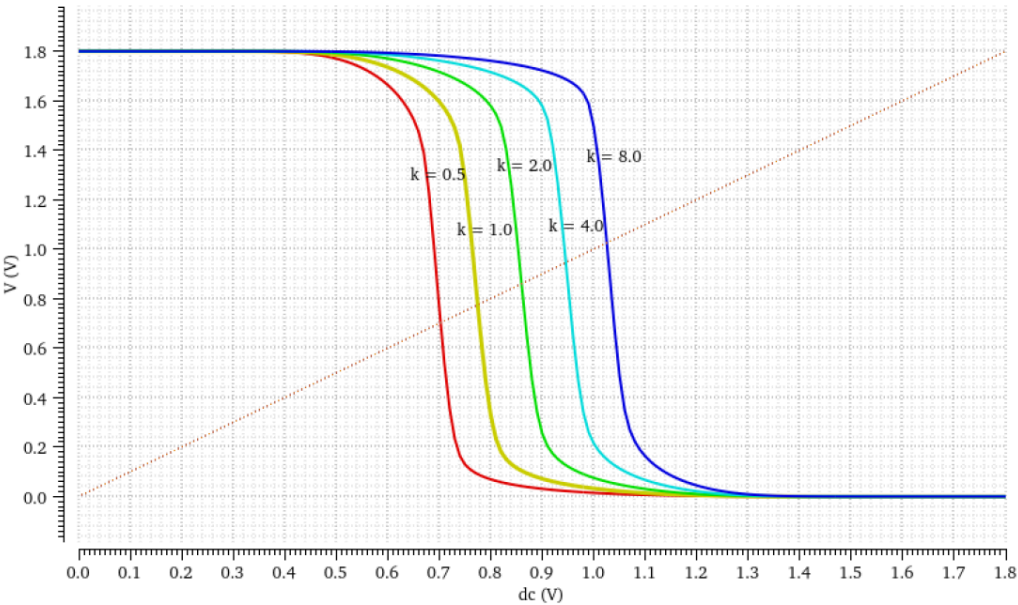


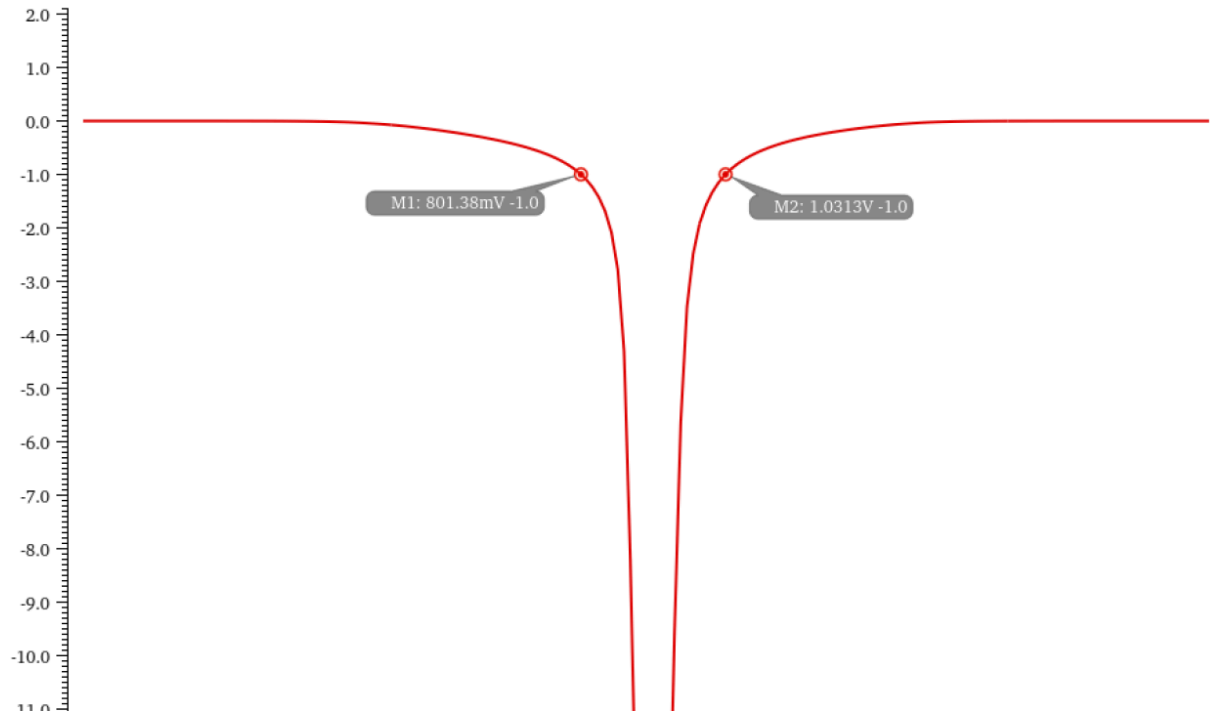
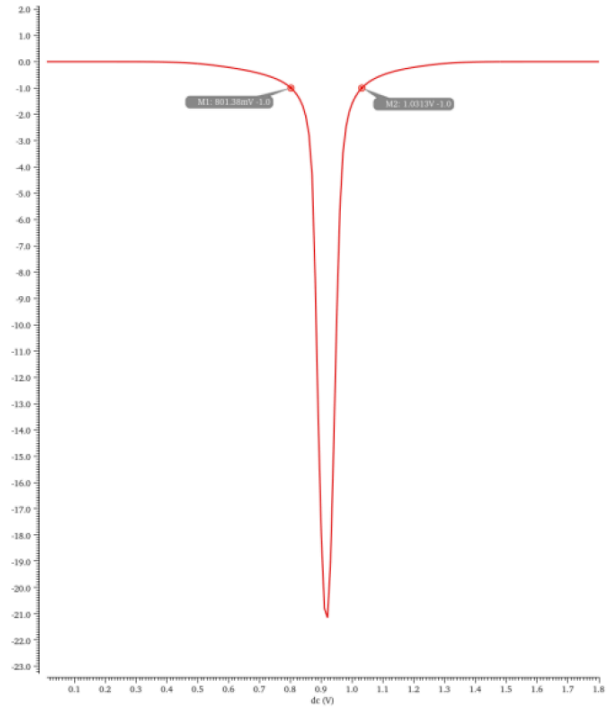
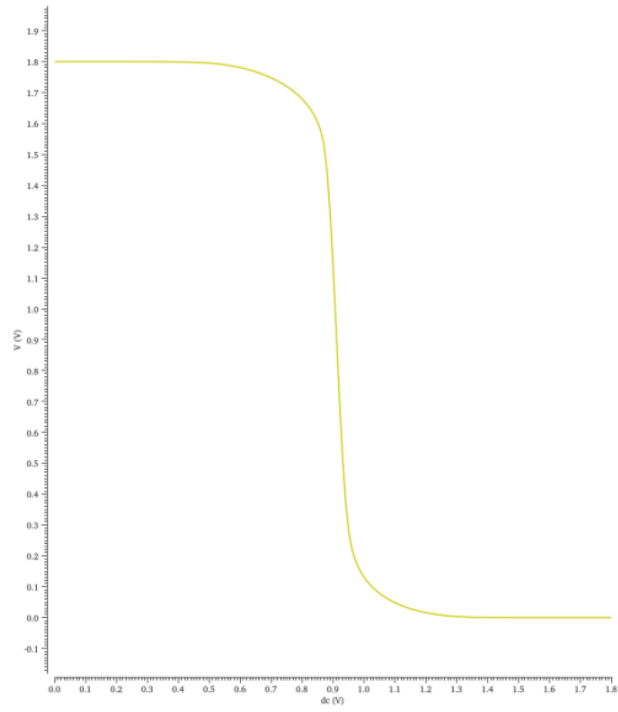
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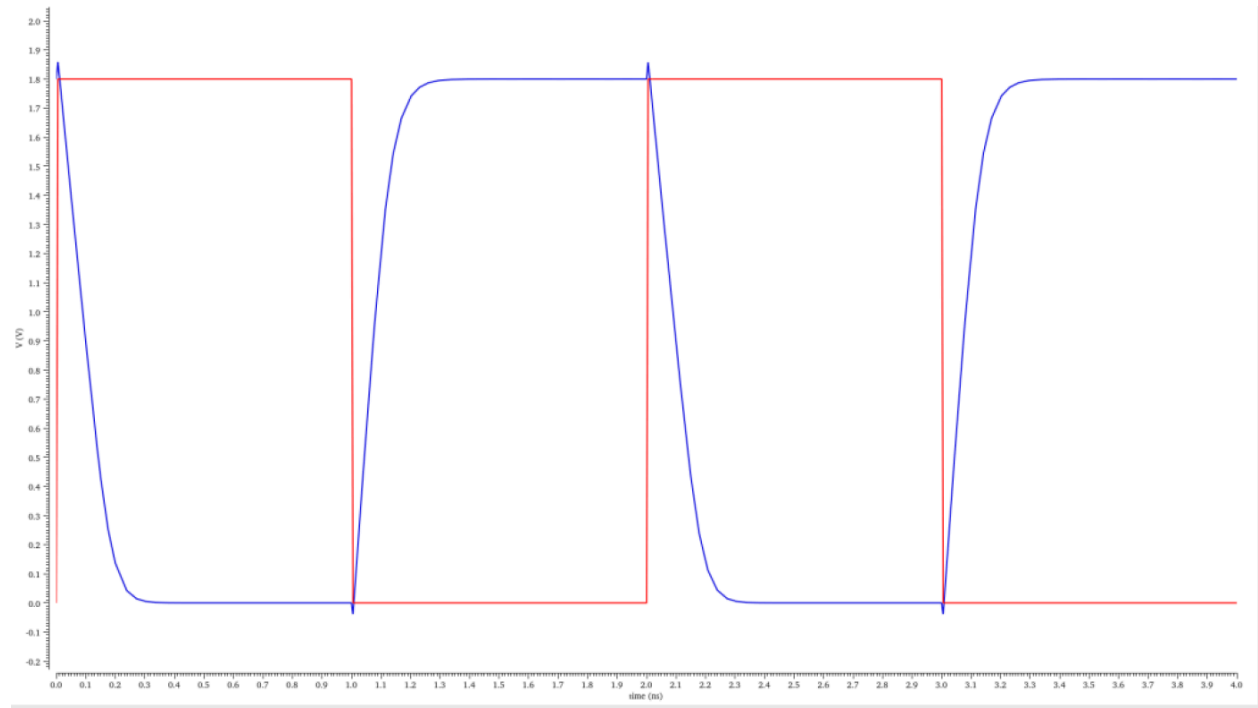
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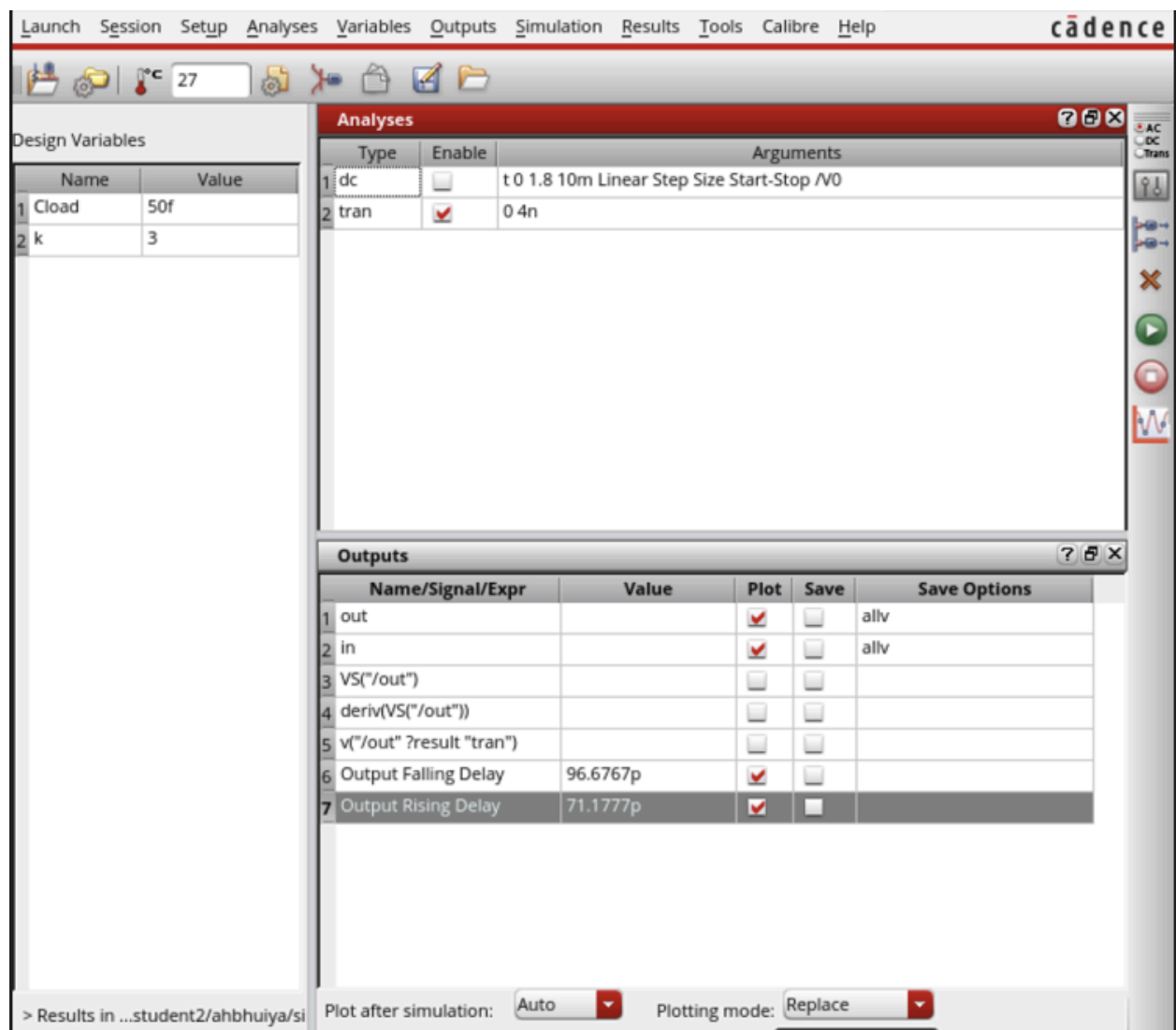
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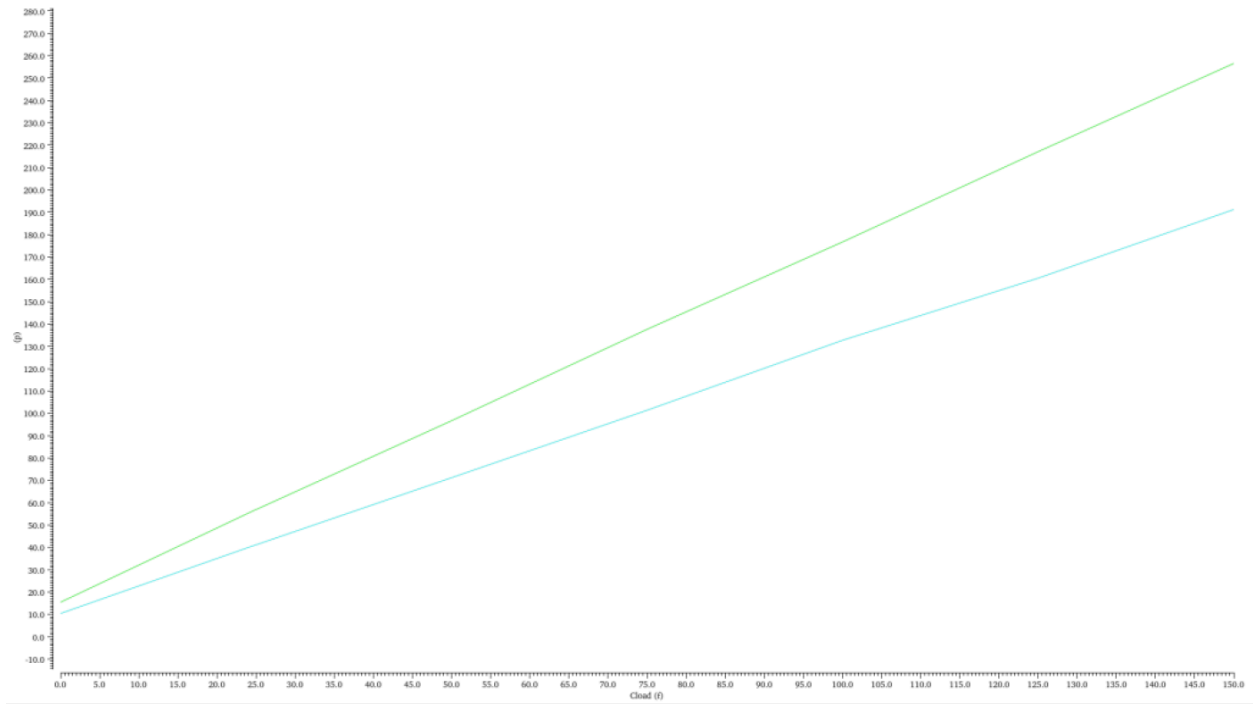
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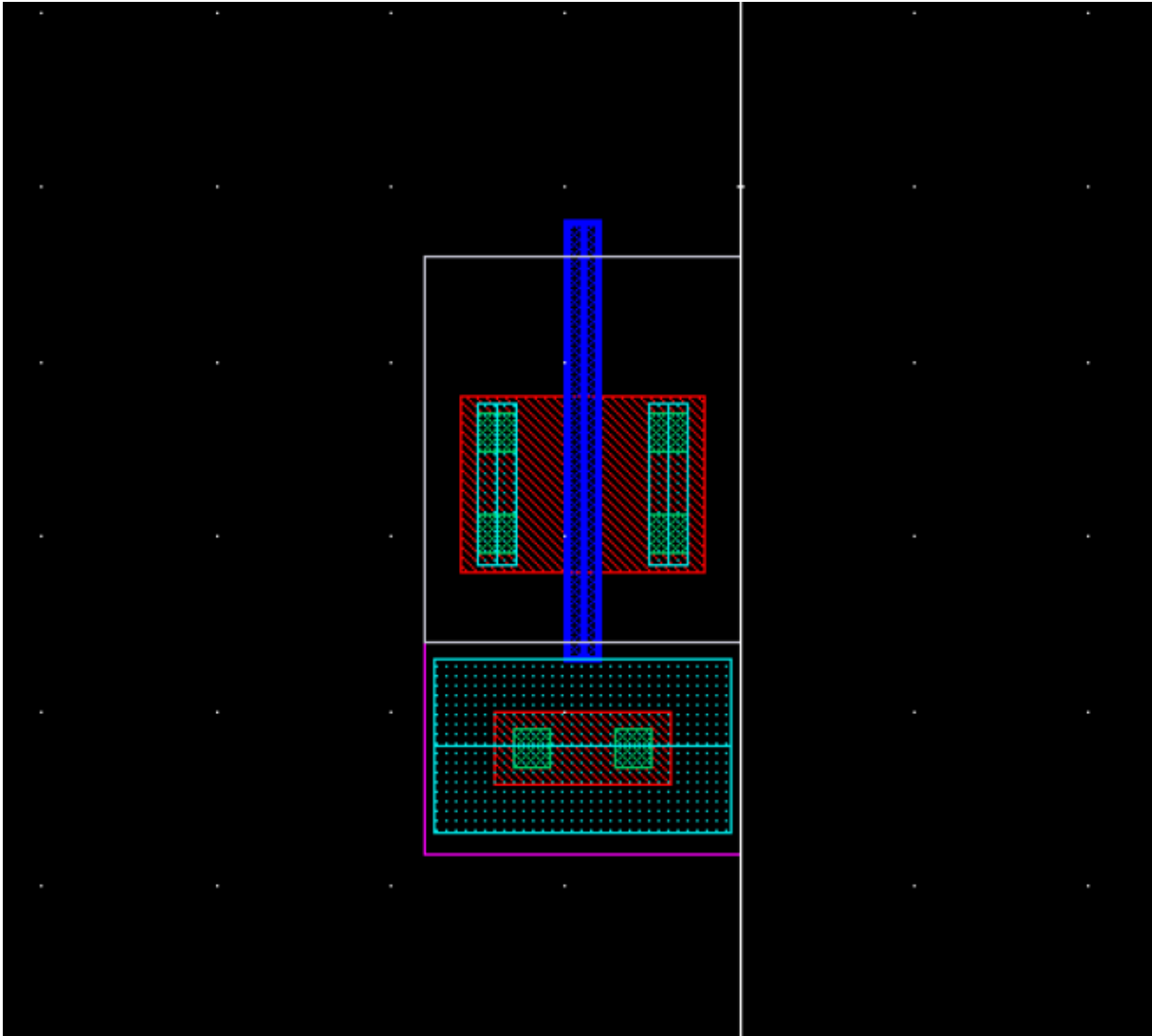


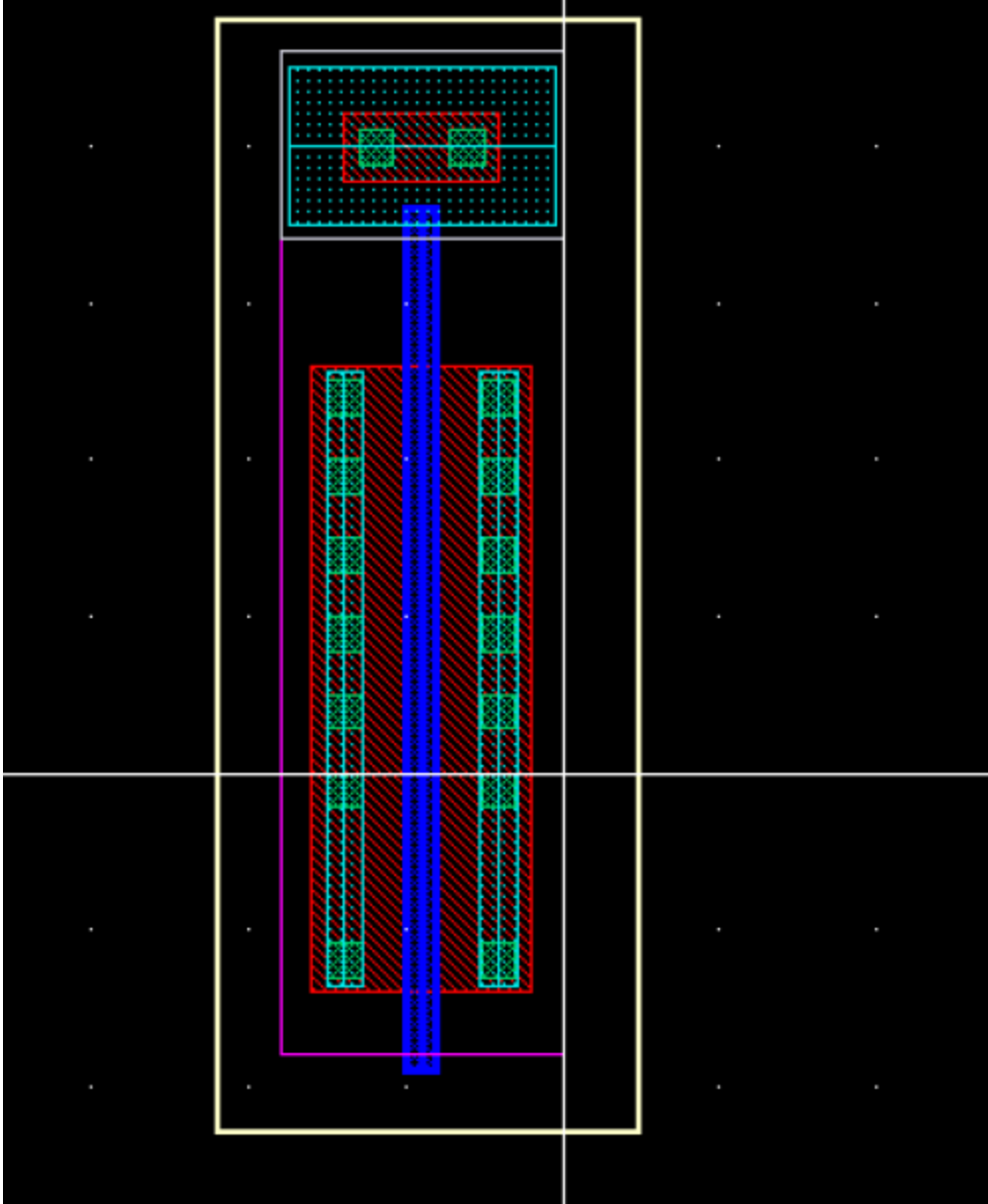


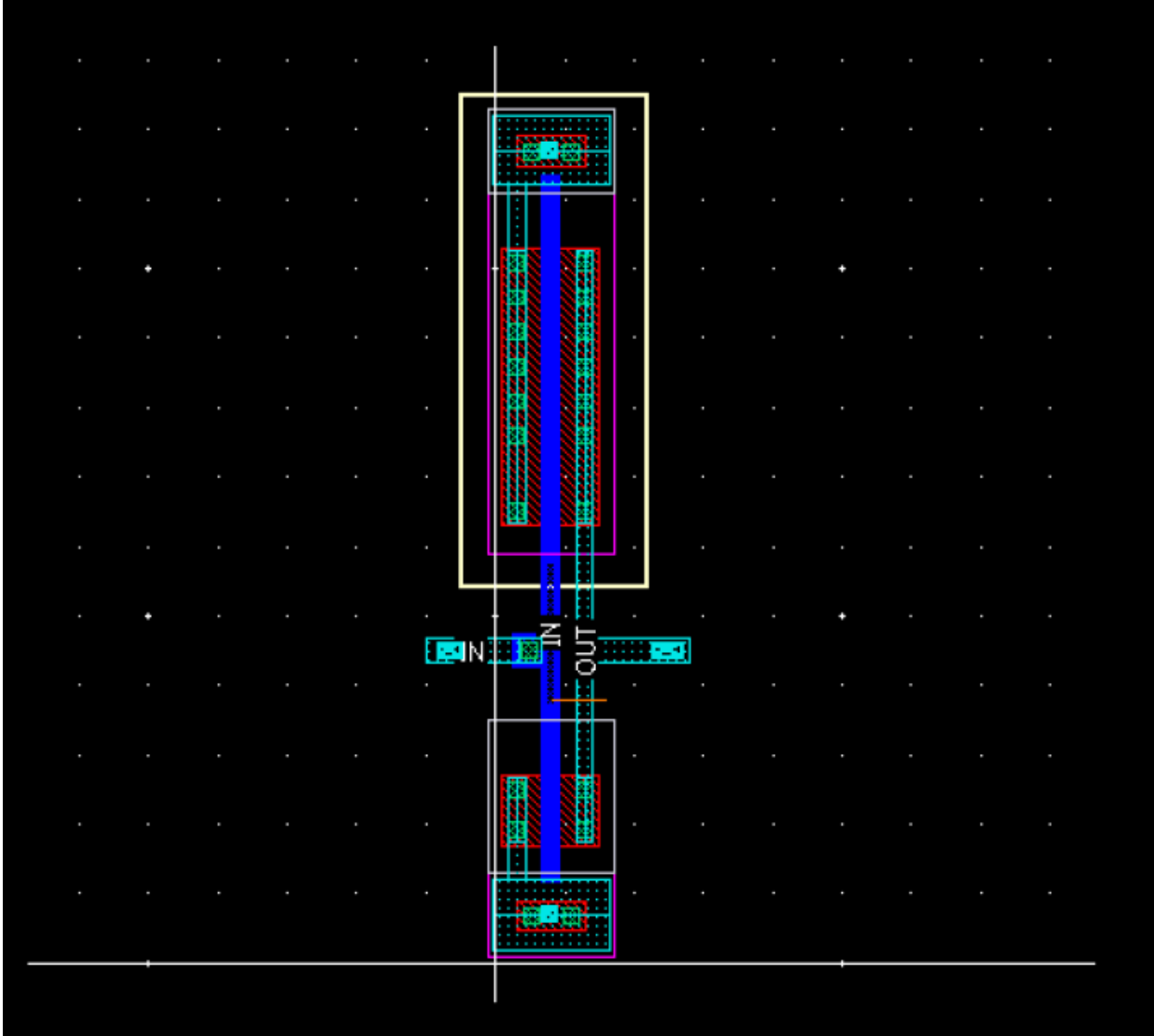




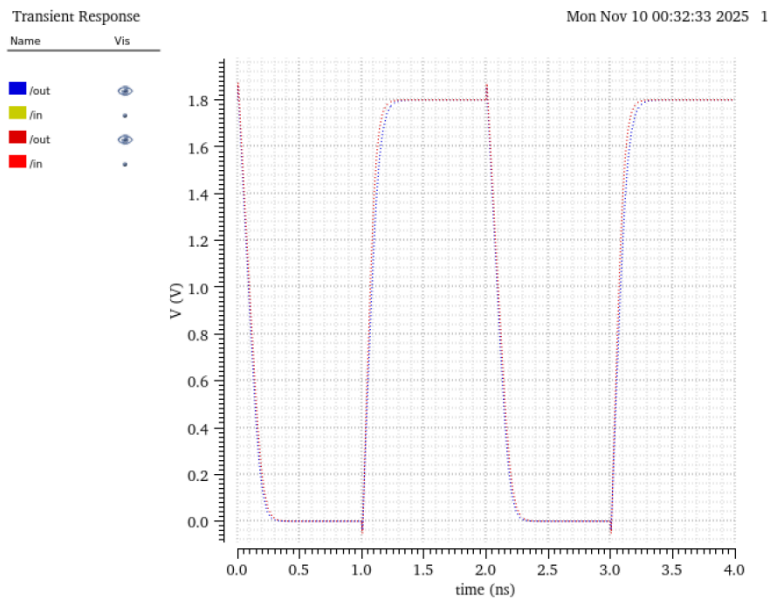
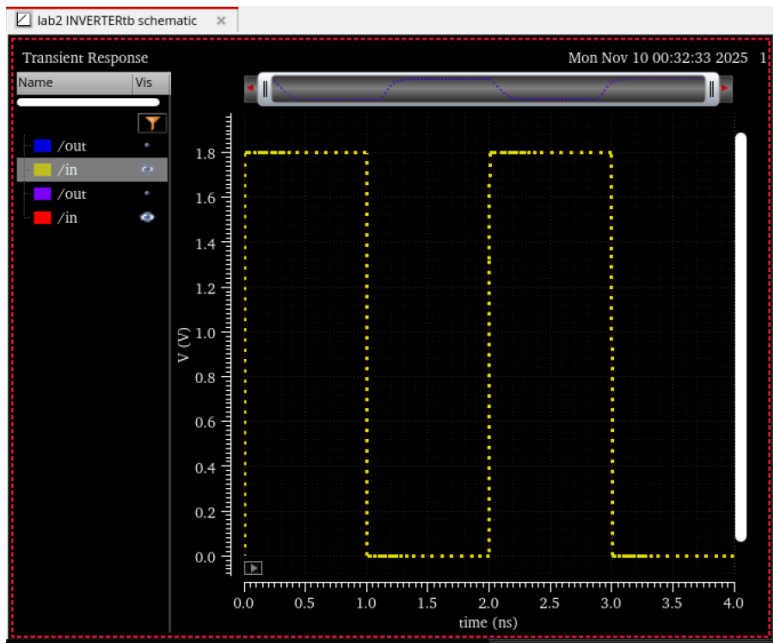


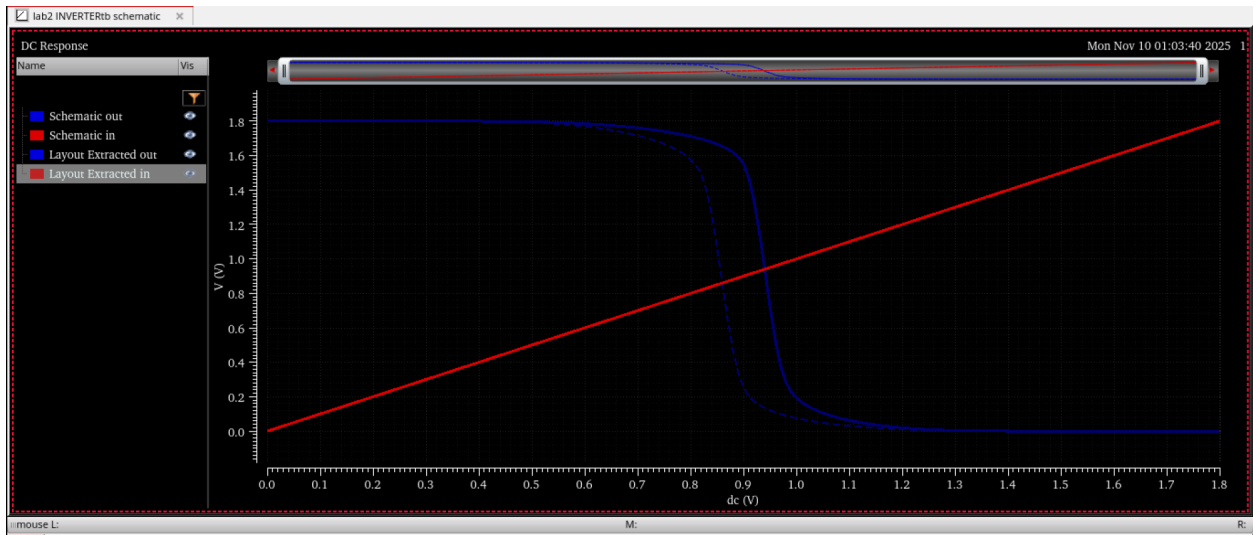
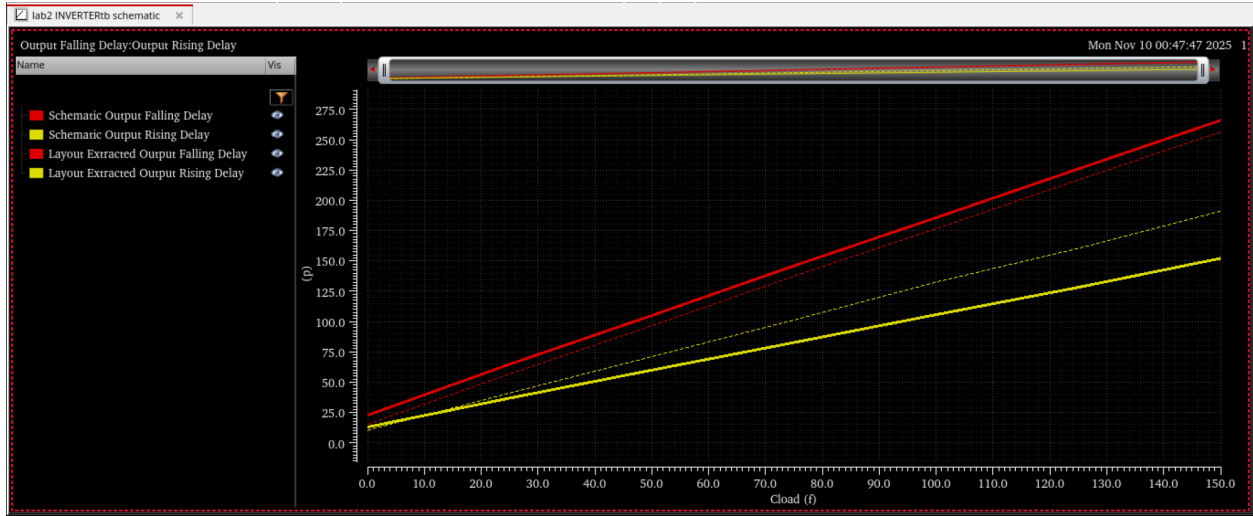






Post-Layout Simulations:





In this lab, the CMOS inverter was designed using theoretical sizing that was explored through from the prelab, where the NMOS transistor was chosen as the unit device and the PMOS transistor was made twice as wide to balance and compensate for the driving strengths of the pull-up and pull-down networks. The prelab analysis, based on the long-channel square-law model, predicted a symmetric voltage transfer characteristic with the switching threshold V_m ideally located at $V_{DD}/2$, along with nearly equal rise and fall delays. However, when the circuit was simulated at the schematic level, and later in the extracted post-layout version, noticeable deviations from the prelab expectations appeared [Note that Post-Layout simulation did not go through as there were previous errors with the NMOS and PMOS layouts/or with the new inverter layout, attempts were made]. The schematic simulation still followed the general inverter behavior, showing a properly defined switching region and quite relatively similar rise and fall times, but the extracted results revealed a clear shift in V_m , slower transitions, and reduced voltage gain in the transfer curve.

These differences arise because the simple prelab model assumed ideal square-law behavior and ignores short-channel effects, parasitics, and real process variations that occur in a real lab environment (Cadence Layout environment in this case). In practice, the electron mobility in the NMOS is significantly higher than the hole mobility in the PMOS, and the PMOS device experiences stronger velocity saturation and series resistance effects. As a result, because of this, PMOS widths that are only twice that of the NMOS does not truly equalize their drive strengths because of this practical variation. The extracted simulation, which includes interconnect capacitances, contact resistances, and other layout parasitics, further exaggerates these imbalances and leads to longer propagation delays and non symmetrical output transitions. In addition to that, effects such as channel length modulation and threshold voltage mismatches contribute to shifting the inverter switching threshold away from the ideal $V_{DD}/2$, in an ordinary balanced case.

In summary, the prelab design did provide a useful theoretical baseline, however real devices exhibit non-ideal behavior that prevents perfect symmetry between rise and fall transitions. The PMOS would need to be sized larger than twice the NMOS, perhaps around three to four times wider, in order to achieve a switching threshold near $V_{DD}/2$ in a modern short-channel process. The discrepancies between the prelab predictions, theoretical simulations, schematic simulations, and extracted results shows the importance of including layout parasitics and realistic device models when designing CMOS circuits for accurate timing and performance.